



Working capital that releases only when the cargo's own evidence says it should.

Evidence-gated trade finance, settled in Paxos USDG on Robinhood Chain.

- **Production build live on Robinhood Chain Testnet. Mainnet release next.**



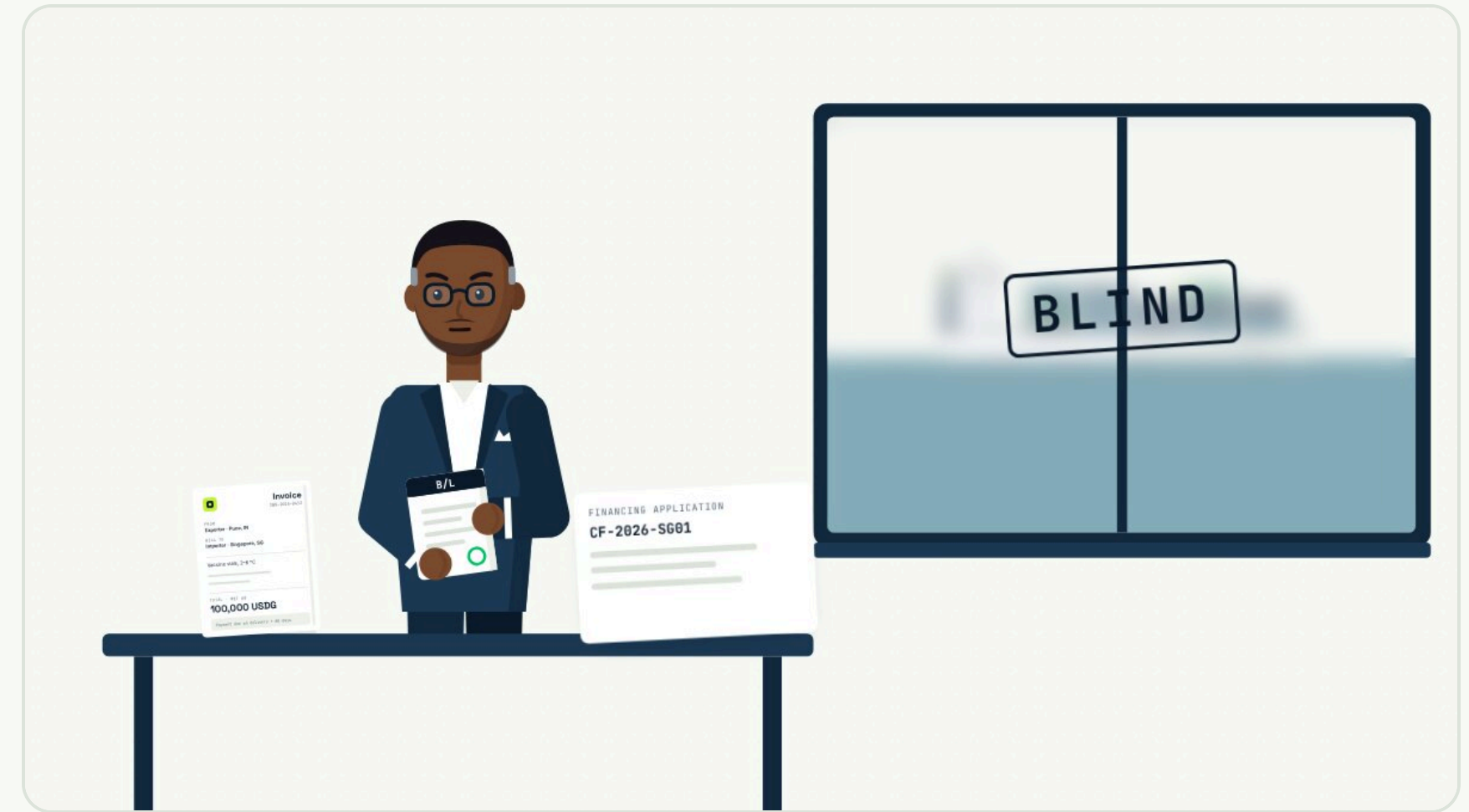
Arbitrum Open House Singapore · Online Buildathon · October 2026

cargoflow.adoranto737.workers.dev · github.com/LSUD0K0/CargoFlow

TRADE FINANCE TODAY

A lender sees paperwork, not the container.

So it lends blind, or does not lend at all. The exporter waits about two months for cash it has already earned, while the cargo is at risk in transit.



Daniel, the financier, holds the documents. The reefer stays out of sight.

SOURCED, HIGHLIGHTED ON THE ORIGINAL PAGES

The gap is large, and it is documented.

\$2.5T

global trade finance gap in 2025 (ADB)

80% of banks polled in the latest [Global Trade Finance Gap Survey](#) released by the Asian Development Bank (ADB).

The report, the only global benchmark of unmet demand for trade finance, estimates **the global trade finance gap remained at \$2.5 trillion in 2025—unchanged from 2023—and represents about 10% of global trade**, down from 10.6%. While this indicates slight progress, the persistently large trade finance gap stifles growth as many firms cannot access the finance they need to maximize trade opportunities.

41%

of SME trade finance applications rejected (ADB)

Small and medium-sized enterprise (SME) financing may be expanding, with more than 80% of banks reporting dedicated strategies for supporting SMEs. For the first time, ADB's Global Trade Finance Gap Survey suggests **SME rejection rates for trade finance (41%) have fallen to nearly the same level as those for large and mid-cap corporates (40%)**, though this requires more research.

Nearly 90% of banks surveyed continue to factor environmental, social, and governance (ESG)

52 days

average B2B payment terms in India (Atradius)

businesses have tightened terms, most have relaxed their payment policies, offering longer terms to support B2B customer relationships. However, this is heightening exposure to customer payment risks. **50% of all B2B sales are currently made on credit, with average payment terms standing at 52 days.**

There is a mixed picture on the working capital management

~\$35B

lost by biopharma each year to cold-chain failures (IQVIA)



© image(s) licensed by DVV Media

The biopharma industry loses approximately \$35 billion annually as a result of failures in temperature-controlled logistics, according to IQVIA Institute for Human Data Science.

WHY NOW

Money became programmable. Now it can answer to evidence.

\$

Paxos USDG

A US-dollar stablecoin from Paxos Digital Singapore. The only money CargoFlow moves.

RH

Robinhood Chain

An Arbitrum chain built for tokenised real-world assets, with USDG deployed.

P-256

Passkeys on chain

The RIP-7212 precompile lets a buyer pay with Face ID, no seed phrase.

Rs

Arbitrum Stylus

Evidence fusion in Rust, measured 19x cheaper than the Solidity reference.

THE SOLUTION

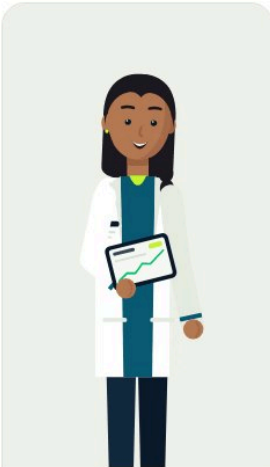
Capital that answers to the cargo's evidence.

- 1

A financier escrows USDG for one shipment.
- 2

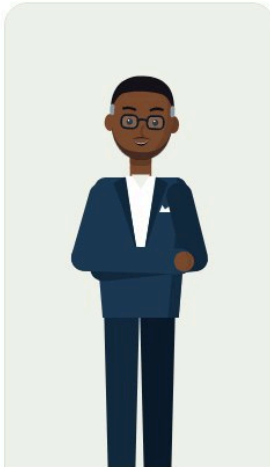
Tranches release only when signed sensor evidence passes an on-chain policy.
- 3

The buyer's one payment repays the financier, pays the exporter and hands over the bill of lading.



Meera
EXPORTER

Registers the shipment and policy, opens the facility, resumes with a proof



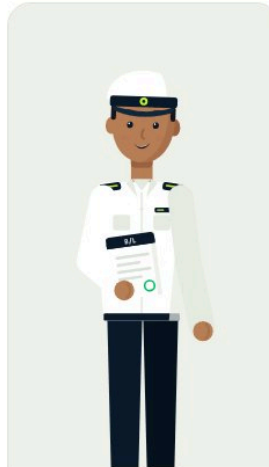
Daniel
FINANCIER

Escrows USDG, watches the evidence, gets principal plus fee



Wei Lin
BUYER

Confirms delivery, pays the invoice, receives the bill of lading



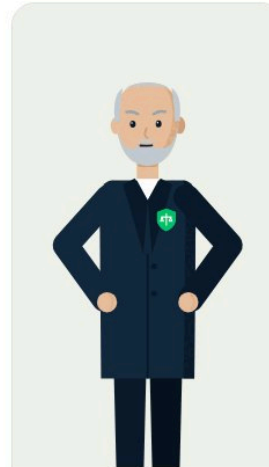
Carrier
SHIP'S OFFICER

Issues the electronic bill of lading (ERC-721)



Insurer
COVER PROVIDER

Offers default cover and parametric cover



Arbiter
DISPUTE ROLE

Resolves disputes; can never release a tranche

Illustrative characters. Everything they do is a real contract call.

HOW IT WORKS

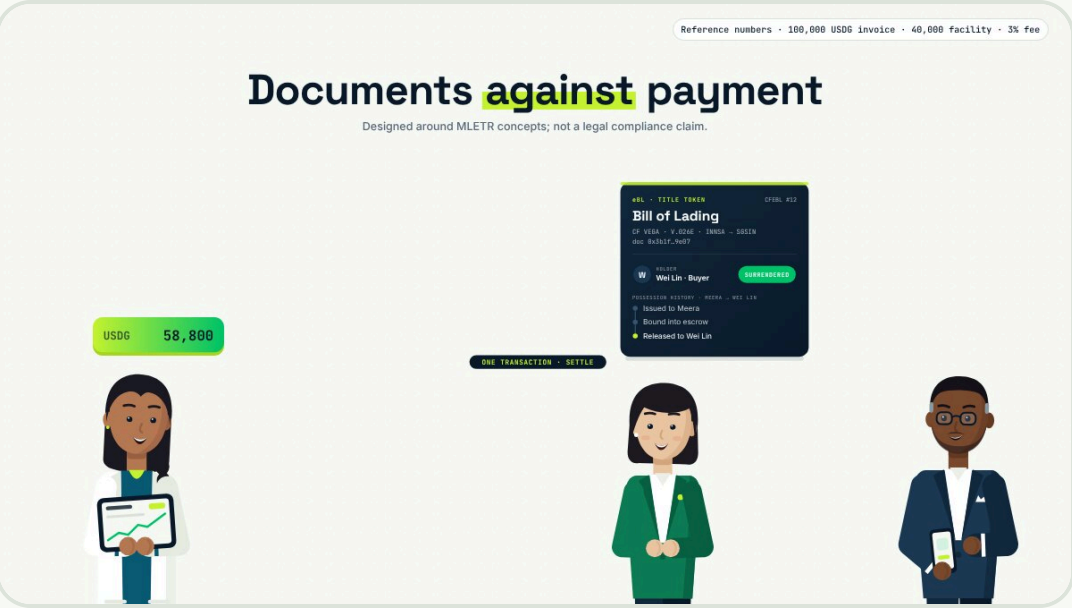
Five steps, every one on chain.



Reference numbers: 100,000 USDG invoice, 40,000 USDG facility, 3% fee.

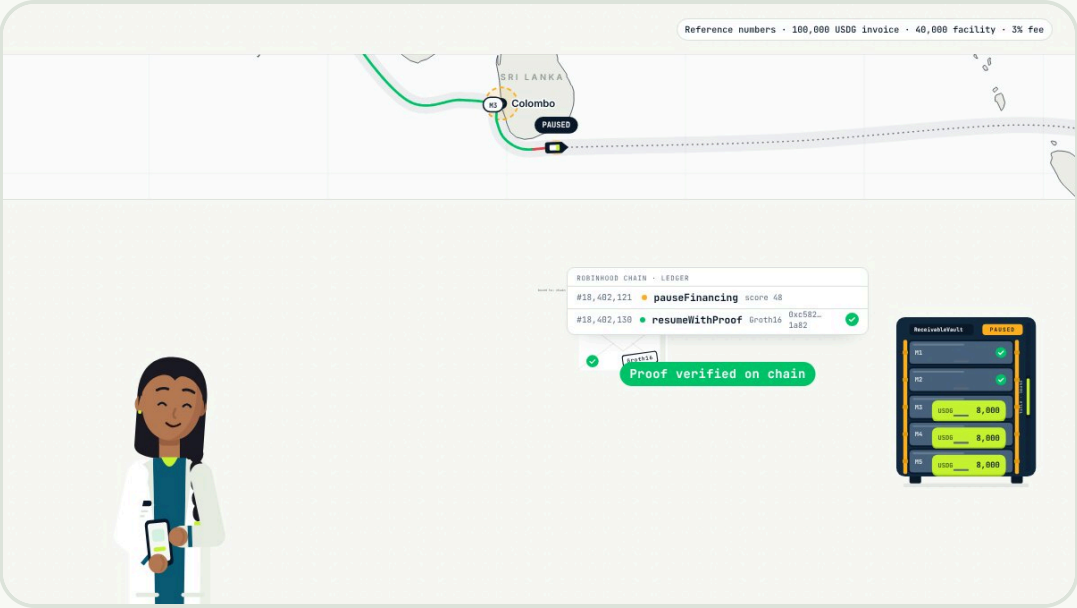
BUILT INTO THE CONTRACTS

Safety nets are part of the protocol.



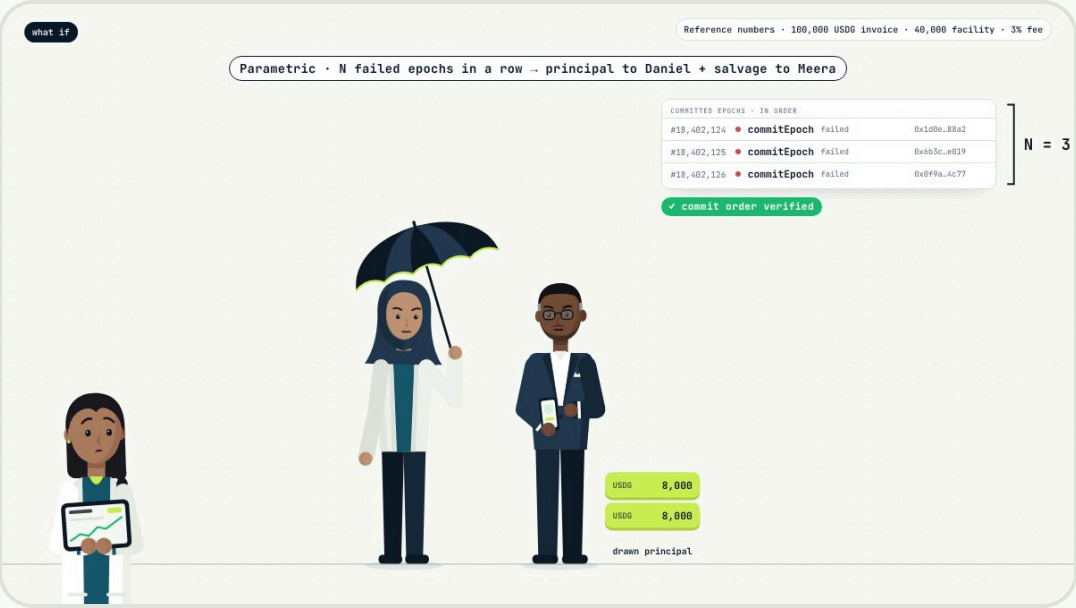
Title moves with money

The carrier's ERC-721 bill of lading leaves escrow inside the buyer's payment.



Automatic recovery

A proof is prepared once a probe has eight fresh in-band readings. The exporter signs once.

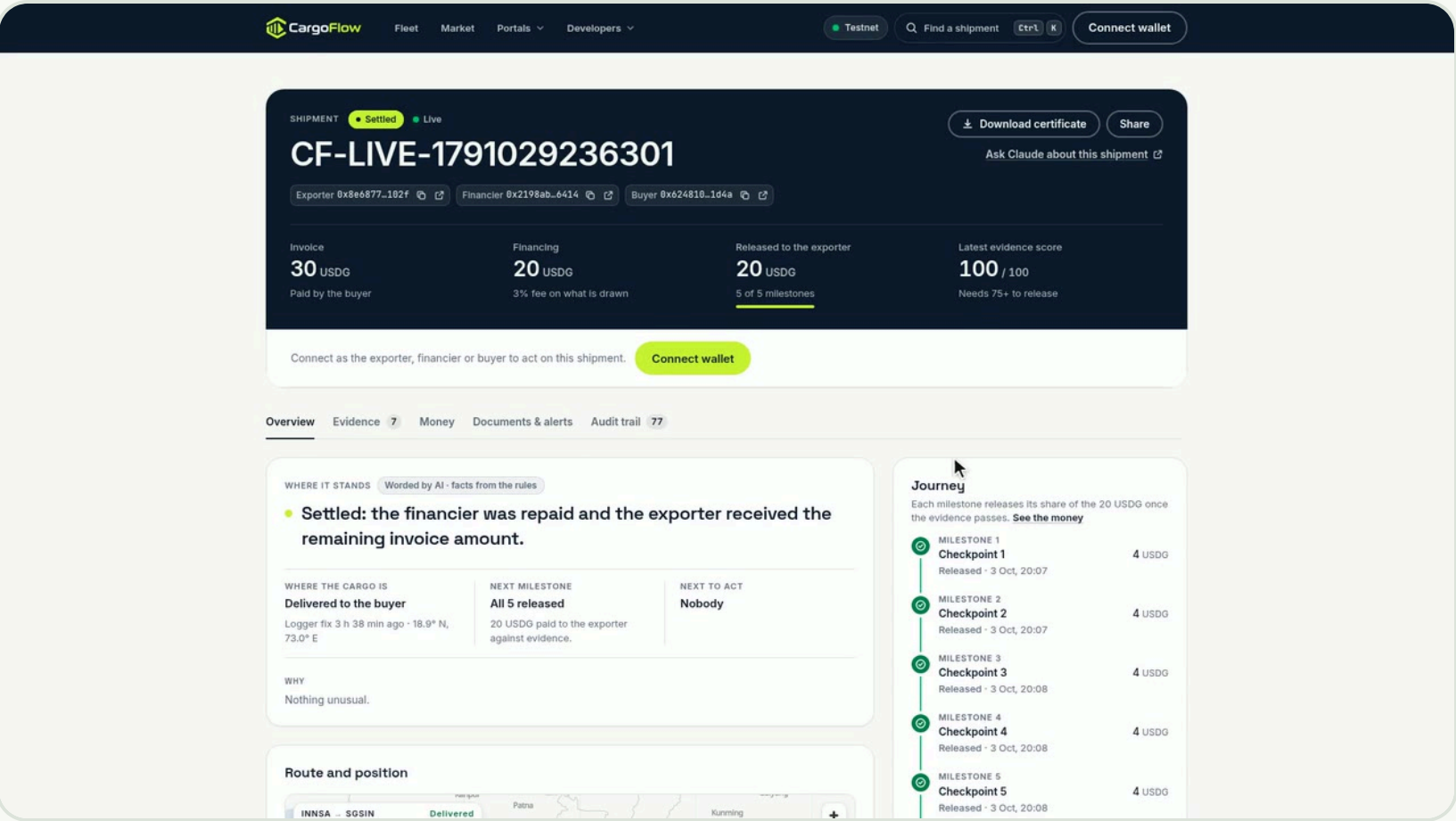


Parametric cover

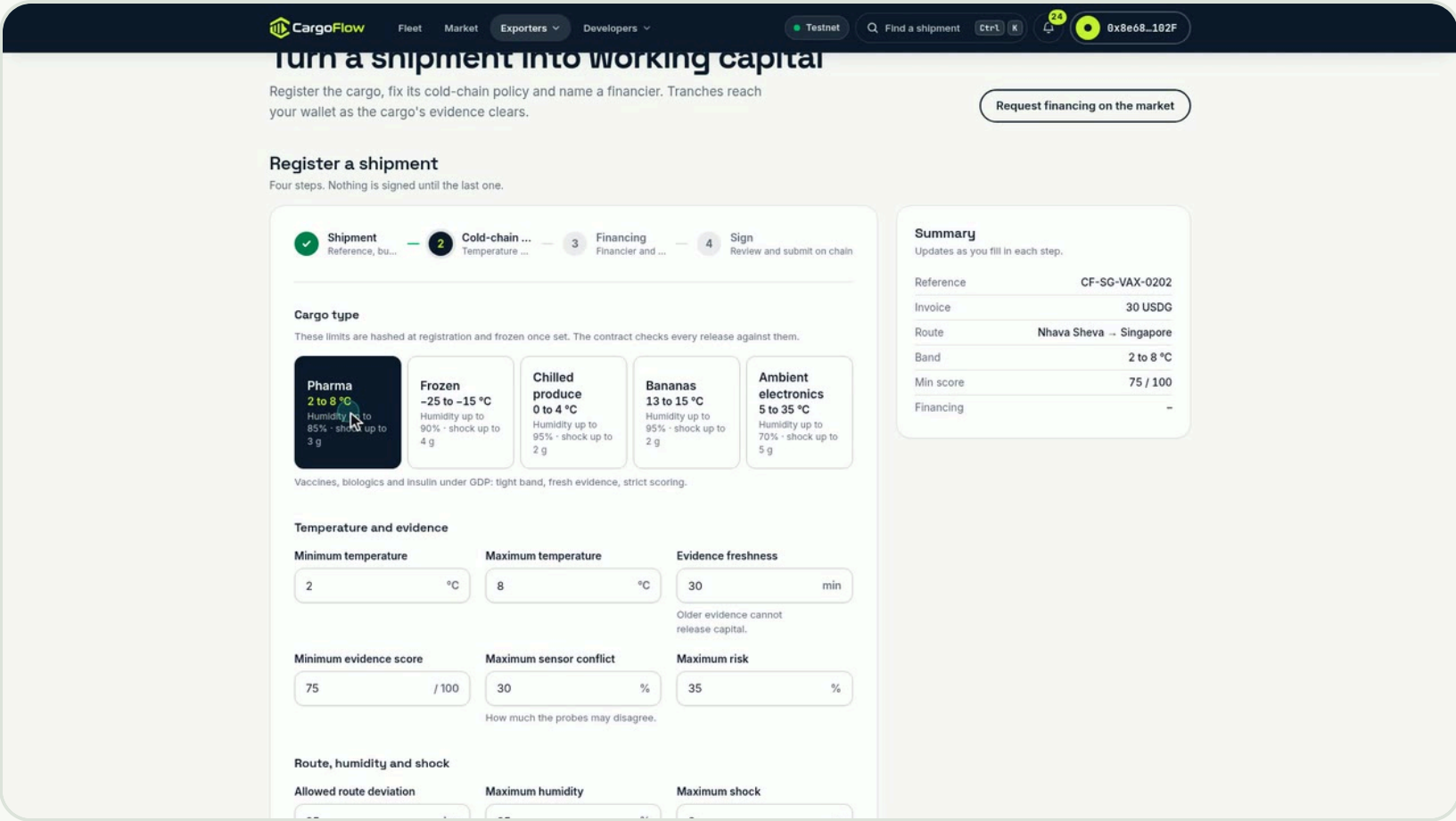
Pays out after N failed epochs in a row, proven from the commit order.

LIVE AT CARGOFLOW.ADORANTO737.WORKERS.DEV

A product each party can use today.



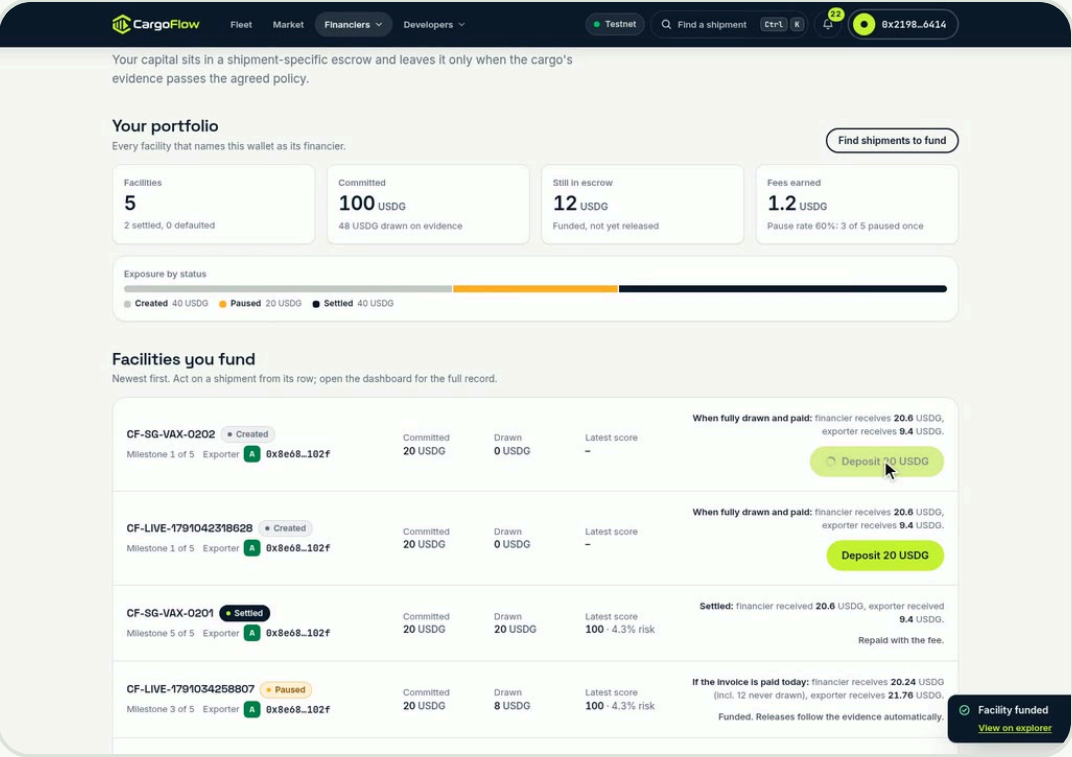
Shipment dashboard: settled, five of five milestones released.



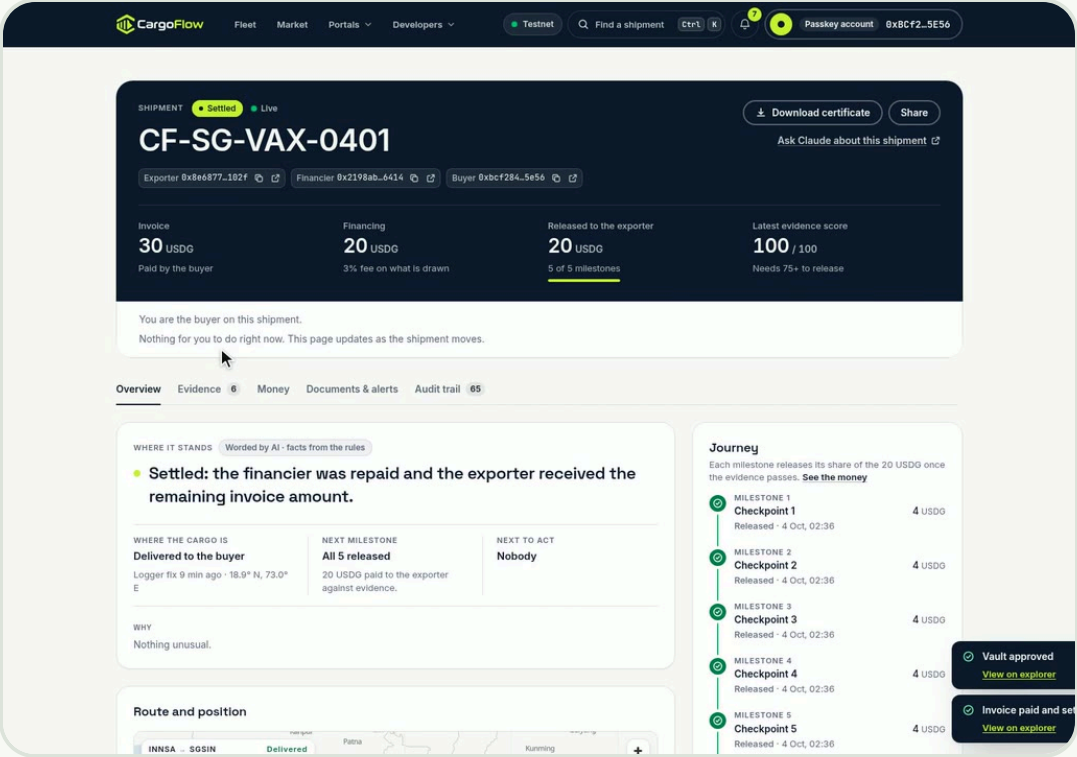
Exporter wizard: the Pharma 2 to 8 °C template fills the policy.

EVERY PARTY, ITS OWN SESSION

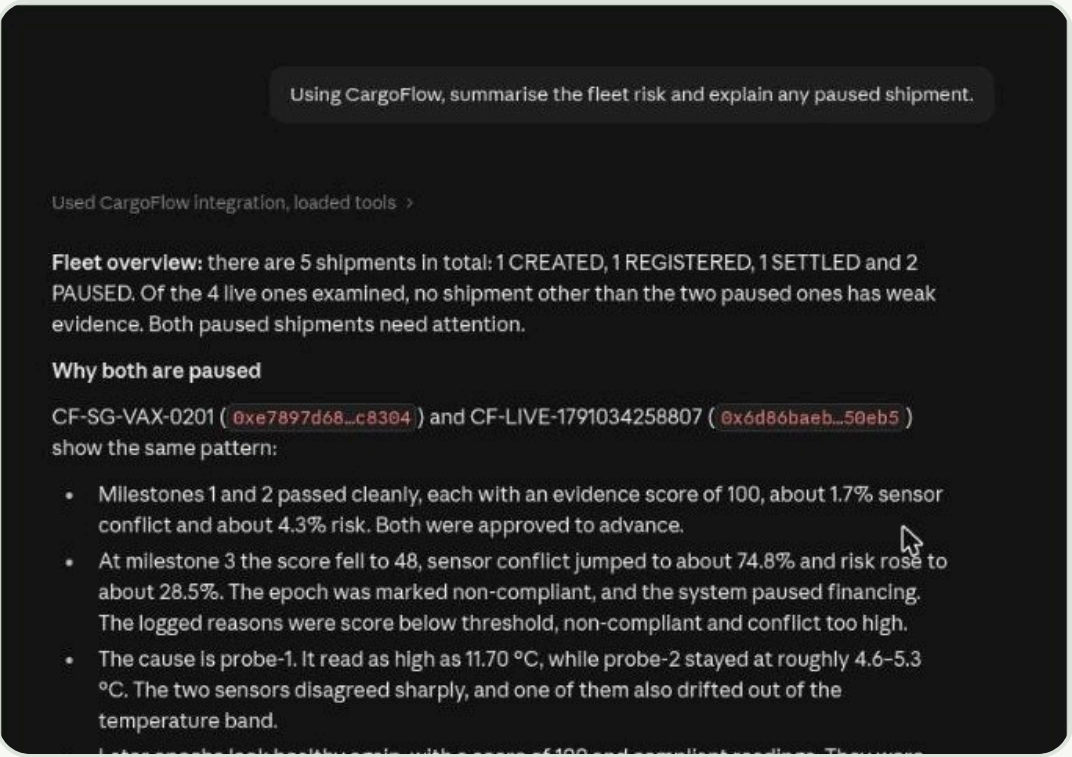
Fund it, pay it, ask Claude about it.



Financier: portfolio, exposure and one-click deposits.



Buyer: confirms delivery and pays with a passkey.

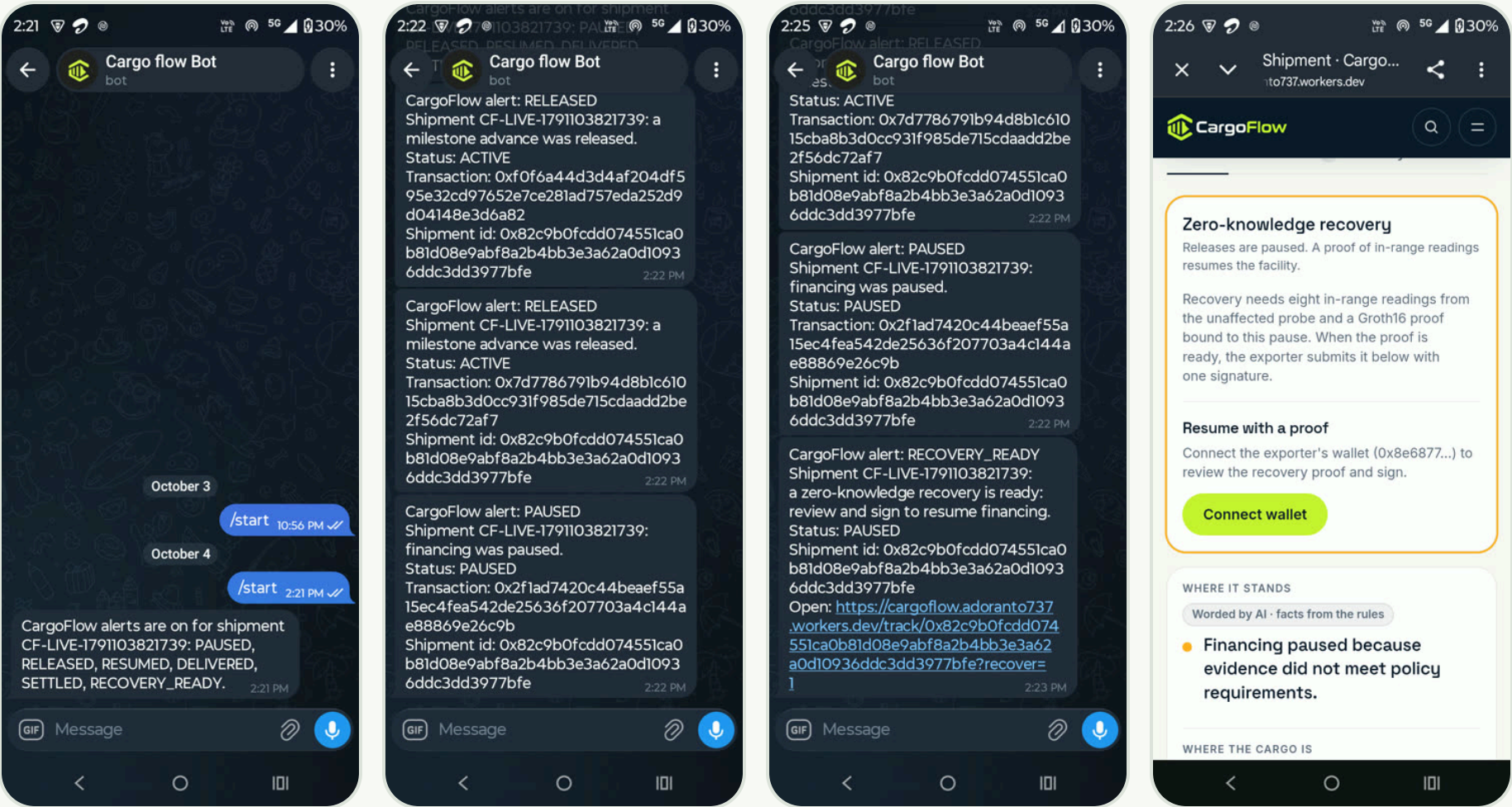


claude.ai: 25 tools through remote MCP. Claude never holds a key.

ALERTS

Alerts on Telegram, email, Slack and webhooks.

Recorded on a real phone during a live testnet run. RELEASED and PAUSED arrive within seconds of the chain events; RECOVERY_READY opens the recovery card, ready to sign.



Alerts on

Released, paused

Recovery ready

One tap to sign

Bot: [@Cargo_FlowBot](#)

TECHNOLOGY AND MOAT

Five mechanisms that enforce each other.

01

Signed readings

Each device signs; every epoch records its source.

02

Fusion score

Dempster-Shafer fusion with a published formula.

03

Bound ZK proof

Groth16, bound to the contract's own signals.

04

Locked vault

Only the controller's rules release USDG.

05

Title with payment

The bill of lading moves inside the payment.

A copy needs all five, tested together: **366** contract tests and **25** circuit tests.

TRACTION

Production build live on Robinhood Chain Testnet.

Mainnet release next.

10

contracts source-verified
on chain

4

packages published on
npm and PyPI

25

MCP tools on a public
remote server

3

extensions on Arbitrum
Sepolia

TESTS, ALL PASSING

366

contract

25

circuit

296

frontend

18

end-to-end

92

SDK

27

MCP

36

gateway

25

Python

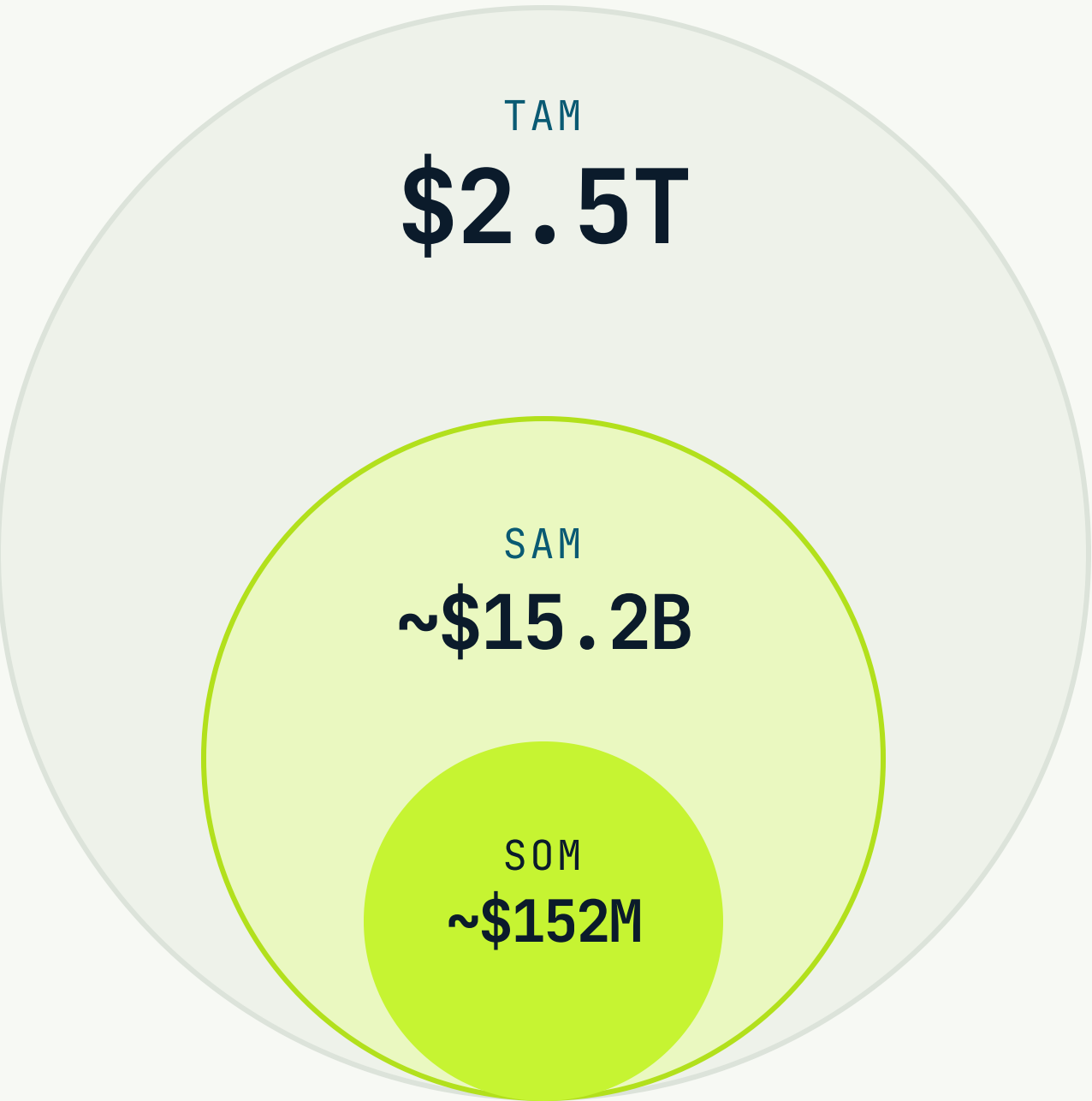
LIVE RUNS

Full lifecycles settled on chain, with receipts.

CF-SG-VAX-0202	Website run: excursion, automatic ZK recovery, settlement.	proof 0x79c68fc0... settle 0xabba66b6...
CF-SG-VAX-0401	The buyer paid with a passkey: three ERC-4337 user operations.	account 0xBCf28437... settle 0xee1d5ba8...
CF-LIVE-1791029236301	v3 reference run through the hosted API, each party signing.	proof 0xadfe3b2f... paid 0x37571b49...
v1 hero run	22 transactions in 116 seconds, proof verified on chain.	proof 0x3910c2a5... settle 0xcb11761c...

MARKET SIZE, BOTTOM-UP

Start with one corridor inside a \$2.5T gap.



TAM \$2.5T	Unmet trade finance demand in 2025 (ADB).	
SAM ~\$15.2B	India's pharma exports, \$30.47B, × 50% sold on credit.	A1
SOM ~\$152M	1% of SAM financed through CargoFlow by year 3.	A2
REVENUE \$0.46–0.76M	A planned 30 to 50 bps fee on that volume, per year.	A3
CAPITAL ~\$22M	Financier capital outstanding at 52-day terms.	A4

Assumptions: A1 Atradius' India-wide B2B credit share applies to pharma · A2 1% is a target · A3 planned fee level · A4 365 / 52, about 7 turns a year · [PIB, FY 2024-25 pharma exports](#) · [ADB](#) · [Atradius](#)

BUSINESS MODEL

Paid when capital moves, not before.

01

Protocol fee

30 to 50 bps of each drawn facility, inside the settlement waterfall (A3).

02

Cover premiums

A share of default and parametric cover written through the CoverPool.

03

Software and data

Portfolio risk, evidence API and MCP for funds and insurers.

ONE 100,000 USDG PAYMENT, SPLIT ON CHAIN (REFERENCE FACILITY)

58,800 to the exporter

40,000 principal

1,200

Principal and the 3% fee go to the financier. The protocol fee is one more line in this split.

LANDSCAPE, BY CATEGORY

Each category solves one piece. CargoFlow
joins them.

	Sees cargo condition	Releases on evidence	Payment and title together	Private telemetry	Stablecoin settlement
Bank trade finance letters of credit, collections	☐	☒	☒	☐	☐
Factoring fintechs advance on the buyer's credit	☐	☐	☐	☐	☐
Cold-chain monitoring loggers and visibility	☒	☐	☐	☐	☐
On-chain RWA credit pooled private credit	☐	☐	☐	☐	☒
CargoFlow a facility per shipment	☒	☒	☒	☒	☒

☒ yes ☒ partly, on documents ☐ not a design goal of the category

GO-TO-MARKET

One corridor first: India pharma to Singapore.



DOOR 1 • LOGGER VENDORS

Make existing loggers finance-ready

@cargoflow/gateway signs and sends logger exports.

DOOR 2 • CREDIT FUNDS

Short-tenor USDG yield, evidence attached

The market suggests a fee band per request.

DOOR 3 • FORWARDERS, INSURERS

Reach exporters through who they use

The Pharma template sets 2 to 8 °C in one click.

ROADMAP, EACH MILESTONE WITH AN EXIT TEST

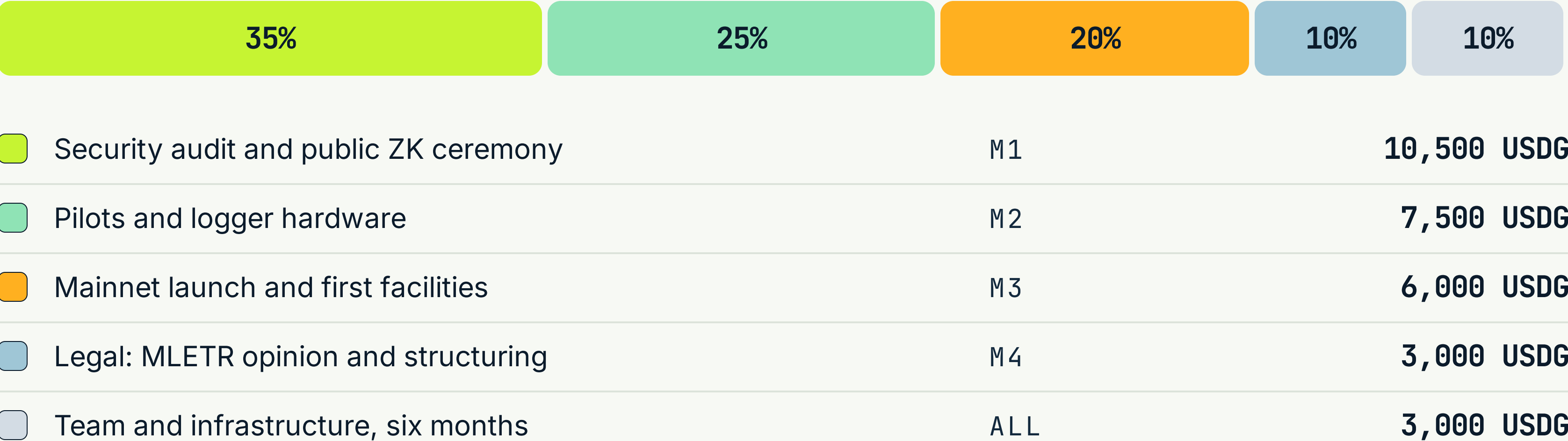
From production build to the first real facility.

M0 done: contracts v3 verified, live app, API and MCP, passkey payments, ZK recovery, four packages.

M1 Security Independent audit and a public ZK ceremony 35% OF GRANT	M2 Real evidence Logger pilots and a secure-element board 25% OF GRANT	M3 Mainnet Real USDG on Robinhood Chain, protocol fee 20% OF GRANT	M4 Legal and title MLETR opinion on the bill of lading 10% OF GRANT	M5 Scale Calibrated scoring, a second corridor AFTER THE GRANT
---	--	--	---	--

THE ASK: THE MILESTONE-BASED GRANT

A 30,000 USDG grant funds M1 to M4.



Released like a CargoFlow facility: each tranche follows a milestone's exit test.

ARBITRUM OPEN HOUSE SINGAPORE

Every criterion, with something to click.

Deployed on an Arbitrum chain

10 contracts verified on Robinhood Chain Testnet (46630).

Product-market fit

A sourced problem, five role portals, four published packages.

Real problem solving

End-to-end runs with every transaction linked.

Robinhood Chain

Its RIP-7212 precompile carries the passkey flow.

Smart contract quality

366 tests, eight invariants, an immutable core with no proxy.

Innovation

Evidence-gated capital, bound ZK recovery, a keyless Claude co-pilot.

Paxos USDG

The only money CargoFlow moves.

[Read the HackQuest listing](#)

TEAM AND CONTACT

Built by its founder, live and open.



Arpit Singh

Founder

github.com/LSUDOKO

WEBSITE cargoflow.adoranto737.workers.dev

GITHUB github.com/LSUDOKO/CargoFlow

NPM [@cargoflow/sdk](https://www.npmjs.com/@cargoflow/sdk) · [mcp](https://www.npmjs.com/mcp) · [gateway](https://www.npmjs.com/gateway)

PYPI pypi.org/project/cargoflow

CLAUDE cargoflow-mcp.adoranto737.workers.dev/mcp